

AMENDMENT NO. 1 JANUARY 1980
TO

IS : 1832-1978 SPECIFICATION FOR
MALATHION, TECHNICAL

(*First Revision*)

Alterations

(*Pages 6, 7, 8 and 9, clauses A-1 to A-1.6.2*) — Substitute the following for the existing clauses:

‘A-1. PRINCIPLE OF METHOD

A-1.1 Malathion, *S*-1, 2-di (ethoxycarbonyl) ethyl *O*, *O*-dimethyl phosphorodithioate, is decomposed by alkali in ethyl alcohol or methyl alcohol to sodium *O*, *O*-dimethyl dithiophosphate sodium fumarate, and ethyl alcohol. The sodium *O*, *O*-dimethyl dithiophosphate is then converted to the cupric complex which is soluble in carbon tetrachloride with the formation of an intense yellow colour. The colour intensity is proportional to the concentration of *O*, *O*-dimethyl dithiophosphoric acid and is measured colorimetrically at 420 nm, the absorption peak. The corresponding amount of the insecticide is derived by comparison with a standard prepared with a known amount of the pure insecticide, which is simultaneously carried through the procedure.

A-1.1.1 In this method, ferric reagent is added to oxidize materials which would reduce cupric ions to cuprous ions; with *O*, *O*-dimethyl dithiophonic acid (DMTA), cuprous ions form a colourless complex which is apparently more stable than the yellow cupric complex.

A-1.2 Apparatus

A-1.2.1 *Spectrophotometer or Photoelectric Colorimeter* — with a blue filter (420 nm) and having 1-cm absorption cell.

A-1.3 Reagents

A-1.3.1 *Malathion*

A-1.3.2 *Acetonitrile* — boiling range 80 to 82°C.

A-1.3.3 *Ethyl Alcohol* — anhydrous, alternatively methyl alcohol anhydrous may be used.

A-1.3.4 Carbon Tetrachloride

A-1.3.5 Sodium Hydroxide Solution — 0.5 N. Dissolve 4.0 g of sodium hydroxide in 200 ml of water to get a clear solution.

A-1.3.6 Ferric Reagent

A-1.3.6.1 Stock Solution — Dissolve 4.0 g of ferric chloride ($\text{FeCl}_3 \cdot 6\text{H}_2\text{O}$) and 160 ml of concentrated hydrochloric acid in water and dilute to 500 ml.

A-1.3.6.2 Dilute solution — Dilute 25 ml of the stock solution with water to 1 000 ml.

A-1.3.7 Copper Sulphate Solution — 1.5 percent (m/v). Prepare by dissolving 1.5 g of copper sulphate ($\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$) in 100 ml of water.

A-1.4 Preparation of Standard and Sample Solutions

A-1.4.0 It is very important in preparing the *Standard* and *Sample Solutions* that volume fluctuations with changing temperatures are avoided. This is accomplished by bringing each solution to the volume mark just before proceeding to the next step in the procedure, as indicated or preferably analysis shall be carried in a room maintained at constant temperature.

A-1.4.1 Preparation of Standard Solution

A-1.4.1.1 Accurately weigh into a 5-ml or 10-ml beaker, 0.100 ± 0.020 g of pure malathion. Quantitatively transfer it to a 100-ml volumetric flask with anhydrous ethyl alcohol or anhydrous methyl alcohol. Then dilute with ethyl alcohol or methyl alcohol to approximately 5 ml below the volume mark. Allow the solution to come to a selected temperature. Record this temperature. Dilute the solution with corresponding alcohol to the mark and mix well. Call this as *Standard Solution A*.

NOTE — *Standard Solution A* may be retained for daily use up to two weeks. It should be preferably kept at low temperature. It is very important, however, that it is adjusted to the temperature recorded above before aliquots are withdrawn.

A-1.4.1.2 Transfer a 10 ml aliquot of *Standard Solution A* into a 100-ml graduated flask. Add 1 ml acetonitrile and dilute with corresponding alcohol to approximately 5 ml below the volume mark. Retain the solution until *Sample Solution* (see A-1.4.2) is prepared and is ready for final dilution. At that time, dilute with corresponding alcohol to the mark and mix well. Call this as *Standard Solution*.

A-1.4.2 Preparation of Sample Solution

A-1.4.2.1 Accurately weigh a quantity of the sample containing 0.100 ± 0.020 g of malathion and quantitatively transfer it to a 100-ml

volumetric flask with anhydrous ethyl alcohol or anhydrous methyl alcohol and make up to the mark with corresponding alcohol at selected temperature (see A-1.4.1.1). Call this as *Sample Solution A*.

A-1.4.2.2 Immediately transfer a 10 ml aliquot of *Sample Solution A* into a 100-ml graduated flask. Add 1 ml acetonitrile and dilute it to mark with corresponding alcohol at selected temperature (see A-1.4.1.1) at the time of final dilution. Call this as *Sample Solution*.

NOTE — If more than one sample solutions are being prepared for assay, then all *Sample Solutions* and the *Standard Solutions* should be diluted to approximately 5 ml below the volume mark as prepared. All shall be diluted to the volume mark at the same time.

A-1.5 Procedure

A-1.5.0 To obtain accurate results, it is important that the analysis of *Standard* and *Sample Solutions* be carried simultaneously through the procedure given in A-1.5.1 to A-1.5.4 without interruption.

A-1.5.1 Transfer a 15 ml aliquot of *Standard Solution* and 15 ml of *Sample Solution* to two different 250-ml separating funnels. Add 2 ml of sodium hydroxide solution and mix well by swirling gently (do not shake) for 5 to 10 seconds. Allow the solution to stand for exactly 2 minutes.

A-1.5.2 Immediately add 75 ml of ferric reagent and mix well by swirling for 10 seconds. Allow the solution to stand for exactly 5 minutes. During this 5 minutes period, fill two 50-ml volumetric flasks to the mark with carbon tetrachloride and after the lapse of 5 minutes simultaneously transfer the carbon tetrachloride from each volumetric flask to the respective separating funnels. Allow the flask to drain for one minute.

NOTE — From this point onward the *Sample Solution* and *Standard Solution* should be handled separately. Carry each one through the next step without interruption.

A-1.5.3 Add 2 ml of copper reagent. Start shaking the separating funnel immediately after addition of copper reagent for exactly one minute. The copper complex of DMTA is unstable in the aqueous phase. The copper reagent should be added with a fast delivery pipette and the shaking should be started with practically no time lapse. Allow the phase to separate. Allow some solvent phase to rinse the stem of the separating funnel, then run the clear solvent phase directly into the cell.

A-1.5.4 Measure the absorbance of the yellow solution within 2 minutes at 420 nm using carbon tetrachloride as the blank.

NOTE — Absorbance measurements of both *Standard* and *Sample Solutions* should be made at the same fixed time after the end of the shaking period. The yellow colour of the copper complex in carbon tetrachloride is unstable, hence the colour should be read within 2 minutes after extraction. During this time the absorbance reading drops very slowly.

A-1.6 Calculation

A-1.6.1 The absorbance measurements of the *Standard* and *Sample Solutions* should agree within ± 10 percent of each other.

A-1.6.2 Malathion content, percent by mass = $\frac{A}{B} \times \frac{m}{M} \times f$

where

A = absorbance of *Sample Solution* at 420 nm,

m = mass in g of pure malathion taken for the preparation of *Standard Solution*,

f = percentage purity of the malathion used for the preparation of *Standard Solution*,

B = absorbance of *Standard Solution* at 420 nm, and

M = mass in g of the material taken for the preparation of the *Sample Solution*.'

(Page 10, clause A-2.3.4) — Substitute the following for the existing clause:

'A-2.3.4 Alcoholic Sodium Hydroxide Solution — 1 N.'

(Page 12, clauses B-3 to B-3.2) — Substitute the following for the existing clauses:

'B-3. DETERMINATION OF PURITY

B-3.1 Preparation of Perchloric Acid — Dissolve 73 g of 70 percent perchloric acid with 124.1 g of acetic anhydride and dilute to 5 litres with glacial acetic acid.

NOTE — If the acetic acid used is not anhydrous, a corresponding larger volume of acetic anhydride will be required, equivalent to the total water content of the other two reagents.

B-3.2 Standardization of Perchloric Acid — Dissolve 0.2 g of anhydrous sodium carbonate, accurately weighed, in 50-ml of glacial acetic acid and titrate this solution with the perchloric acid solution, using 1 percent α -naphtholbenzein solution in benzene as indicator. 0.200 g of sodium carbonate corresponds to 37.73 ml of 0.1 *N* perchloric acid.

B-3.3 Determination of Purity of the Sample — Dissolve 0.300 g of the salt in 25 ml of glacial acetic acid and titrate against 0.1 *N* perchloric acid, potentiometrically (see B-3.2).

B-3.4 Calculate the percentage purity of the salt. The purity obtained by the method is greater than 99 percent.'

(AFCD 6)